

ROSEMARY AGDM, AB, Canada

WMO: 715370

Lat: 50.8344N

Lon: 112.0573W

Elev: 729

StdP: 92.87

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-32.6	-28.8	-35.9	0.1	-32.5	-31.9	0.2	-28.6	12.0	-1.6	10.9	0.0	2.4	250	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	30.9	17.7	28.9	17.0	27.1	16.5	19.5	27.7	18.4	26.4	17.4	25.0	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.9	23.4	15.4	11.9	21.9	14.3	11.1	20.6	58.9	27.3	55.1	26.4	51.9	25.3	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	9.8	8.4	-37.9	34.3	4.0	1.6	-40.7	35.5	-43.0	36.4	-45.3	37.3	-48.1	38.5		
			-38.0	22.3	4.1	0.8	-41.0	22.9	-43.4	23.4	-45.6	23.8	-48.6	24.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-9.8	-10.2	-2.8	4.5	10.9	14.9	18.1	16.6	11.6	4.4	-3.6	-10.8	(6)
	DBStd	12.05	8.71	8.44	8.01	4.98	4.11	2.94	2.77	2.97	4.21	4.99	7.51	8.51	(7)
	HDD10.0	3057	614	566	398	171	38	2	0	1	30	182	408	646	(8)
	HDD18.3	5387	873	799	656	415	231	109	38	70	203	431	657	904	(9)
	CDD10.0	767	0	0	1	6	67	149	251	204	78	9	1	0	(10)
	CDD18.3	56	0	0	0	0	1	7	31	16	2	0	0	0	(11)
	CDH23.3	1616	0	0	0	9	111	192	656	469	172	7	0	0	(12)
	CDH26.7	474	0	0	0	1	22	44	214	142	51	0	0	0	(13)
	WSAvg	3.7	3.6	3.3	3.9	4.3	4.3	4.0	3.5	3.3	3.5	3.8	3.7	3.4	(14)
	PrecAvg	350	12	11	17	30	53	71	43	39	29	17	16	12	(15)
(16) Precipitation	PrecMax	534	26	24	57	68	92	201	102	94	110	47	38	28	(16)
	PrecMin	159	1	1	1	8	13	23	7	1	2	3	1	1	(17)
	PrecStd	67	6	6	11	15	22	38	26	22	23	11	10	7	(18)
	PrecStd	67	6	6	11	15	22	38	26	22	23	11	10	7	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	10.4	17.8	24.3	29.3	30.8	33.6	32.5	31.4	24.2	17.8	9.9	(19)
		MCWB	4.4	4.8	8.6	11.1	14.6	18.5	19.0	17.4	16.1	12.2	9.4	4.6	(20)
	2%	DB	6.6	6.8	14.6	20.7	26.2	27.7	31.2	30.2	27.8	20.1	13.0	6.4	(21)
		MCWB	2.8	2.8	6.8	9.7	13.3	16.3	19.1	17.2	15.1	10.8	6.5	2.9	(22)
	5%	DB	4.3	3.9	11.5	17.6	23.8	25.4	29.0	28.1	24.9	17.4	9.6	4.0	(23)
		MCWB	1.7	1.3	5.3	8.1	12.7	15.5	18.1	17.0	14.0	9.5	4.4	1.5	(24)
	10%	DB	2.2	1.3	8.0	14.9	21.2	23.4	27.1	26.0	22.1	14.3	6.9	1.5	(25)
		MCWB	0.2	-0.3	3.5	6.8	11.2	14.5	17.4	16.2	12.8	7.8	2.8	-0.5	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	5.0	9.3	11.9	15.5	19.4	21.5	20.3	17.4	13.4	9.4	5.2	(27)
		MCDB	8.9	9.0	16.8	22.8	26.0	27.4	29.5	28.2	27.8	22.0	17.4	8.8	(28)
	2%	WB	3.2	2.6	7.4	9.9	14.2	17.9	20.1	18.8	15.7	11.3	6.8	3.0	(29)
		MCDB	6.0	5.6	13.6	19.8	24.5	25.7	28.3	26.9	25.7	18.9	12.5	5.8	(30)
	5%	WB	1.9	1.3	5.4	8.5	13.1	16.5	19.0	17.7	14.5	9.7	5.0	1.4	(31)
		MCDB	4.3	3.5	10.8	16.4	22.2	23.3	27.3	25.9	23.8	16.7	9.2	3.9	(32)
	10%	WB	0.4	-0.6	3.6	7.1	11.9	15.3	18.0	16.7	13.2	8.2	3.1	-0.5	(33)
		MCDB	2.3	1.1	7.4	13.7	19.8	21.8	25.4	24.5	21.4	13.8	6.2	1.3	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	12.1	13.1	13.3	14.9	15.8	14.1	16.0	16.4	16.8	15.4	13.1	12.1	(35)
		MCDBR	13.3	15.4	17.9	21.7	21.0	18.3	20.0	21.4	23.2	20.5	16.9	14.1	(36)
		MCWBR	10.1	11.4	10.6	11.3	9.6	8.7	9.2	10.0	11.0	11.5	11.0	10.6	(37)
	5% WB	MCDBR	12.8	14.5	16.8	20.1	18.9	16.7	18.1	18.9	21.7	19.5	16.4	14.1	(38)
		MCWBR	10.0	11.3	10.5	10.8	8.9	8.7	9.0	9.5	10.6	11.3	11.0	10.9	(39)
(40) Clear-Sky Solar Irradiance	taub		0.271	0.285	0.298	0.350	0.376	0.366	0.364	0.383	0.334	0.293	0.264	0.254	(40)
	taud		2.375	2.414	2.444	2.302	2.278	2.357	2.392	2.310	2.428	2.520	2.485	2.395	(41)
	Ebn at Noon		770	860	912	890	879	889	885	846	858	842	785	740	(42)
	Edh at Noon		61	77	90	117	126	117	112	115	92	69	54	51	(43)
(44) All-Sky Solar Radiation	RadAvg		1.18	2.18	3.54	4.85	5.80	6.23	6.74	5.43	3.87	2.37	1.28	0.90	(44)
	RadStd		0.10	0.20	0.26	0.35	0.39	0.38	0.32	0.36	0.35	0.27	0.11	0.08	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (82 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page